| For use by the Project lecturer | Approved | ✓ | Revision required | |
|---|---|---|---|---|

**Feedback**

**For use by the Project module lecturer only**

Approved - the project is clearly delineated.
However, a number of things to pay attention to in especially section 4.

Symbol awarded: B

To be completed by the student

# PROJECT PROPOSAL 2026

| Project no | HCM4 | Revision no | 0 |
|---|---|---|---|

| Title | Surname | Initials | Student no | Study leader (title, initials, surname) |
|---|---|---|---|---|
| Mr | **Braune** | **D.R** | 23539195 | Prof. H.C. Myburgh |

Project title (this is the title on the project concept note)

**Handwriting robot**

| Language editor details | Language editor signature |
|---|---|
| M.T. Braune | MTBraune |
| **Student declaration**<br>I understand what plagiarism is and that I have to complete my project on my own. | **Study leader declaration**<br>This is a clear and unambiguous description of what is required in this project. Approved for submission (Yes/No) |
| Student signature | Study leader signature and date<br>20 April 2026 |

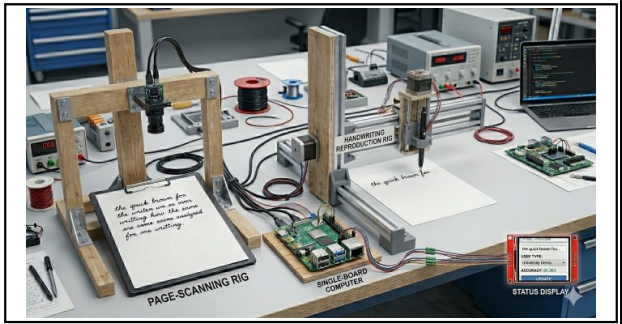
## 1. Project description.

What is the problem to be solved with your project? Describe in non-technical language what your project is about. What does your system have to do? Refer explictly to Figure 1 that you inserted on the right.

The need for handwriting still exists for authenticity, ease of communication and personal touches. The problem is that many individuals do not possess the time, skills or ability to write neatly and legibly. This leads to the solution presented in this proposal: a handwriting robot capable of recognizing a user's handwriting style and being able to replicate it.

The proposed solution is a handwriting robot capable of training on multiple different user's handwritten text (using a page-scanning rig to capture the handwritten page like the one seen in the image) and being able to produce any desired piece of text in any of these user's writing styles (using a text reproduction rig like the one seen in the image). The system will be able to recognize which user authored a piece of text and identify what text was written (shown to the user through a display as seen in the image). A gantry with 2-dimensional movement will perform actions with a pen to automatically reproduce the desired text.

**Figure 1.** Insert an image below that captures the idea with your project. Explain in the text on the left.



## 2. Technical challenges in this project

Describe the technical challenges that are *beyond* those encountered up to the end of third year and in other final year modules.

### 2.1 Primary *design* challenges

**What is the intended core technical challenge in the project? Which aspect or aspects of the design of the system do you expect to be the most challenging?**

1. Text image processing will be performed from first principles, including the isolation of letters from an image of various fonts and sizes of handwritten text.
2. A neural network will be designed and trained from first principles to be able to categorize text to different user's styles based on certain style elements, and identify the text written.
3. A gantry rig will be designed to be capable of performing the necessary precise 2-dimensional movements using a pen, whilst holding the pen and paper securely in place.
4. Pen path planning will be implemented, which involves the conversion of written characters to a 2D motion the gantry needs to perform, and then mapping the movement to stepper motor controls.

### 2.2 Primary *implementation* challenges

**Which aspect or aspects of the implementation do you expect to be the most challenging?**

1. Image processing and deep learning will be performed on a single board computer (instead of on a personal computer), which has limited memory, storage and power.
2. Smooth motor control will be needed to produce precise and consistent pen movement to create accurate text.
3. Near real-time classification of handwritten text and reproduction of digital text on paper will need to be achieved.

All processing will be done on an embedded platform, and not on a PC platform. Confirmed by having ticked this box. ☑

Library functions will be used for hardware interfacing only. Confirmed by having ticked this box. ☑

## 3. Functional analysis

### 3.1 Functional description

Describe the design in terms of system functions as shown on the functional block diagram in section 3.2. This description should be in *narrative format*. **DO NOT** use a bullet list.

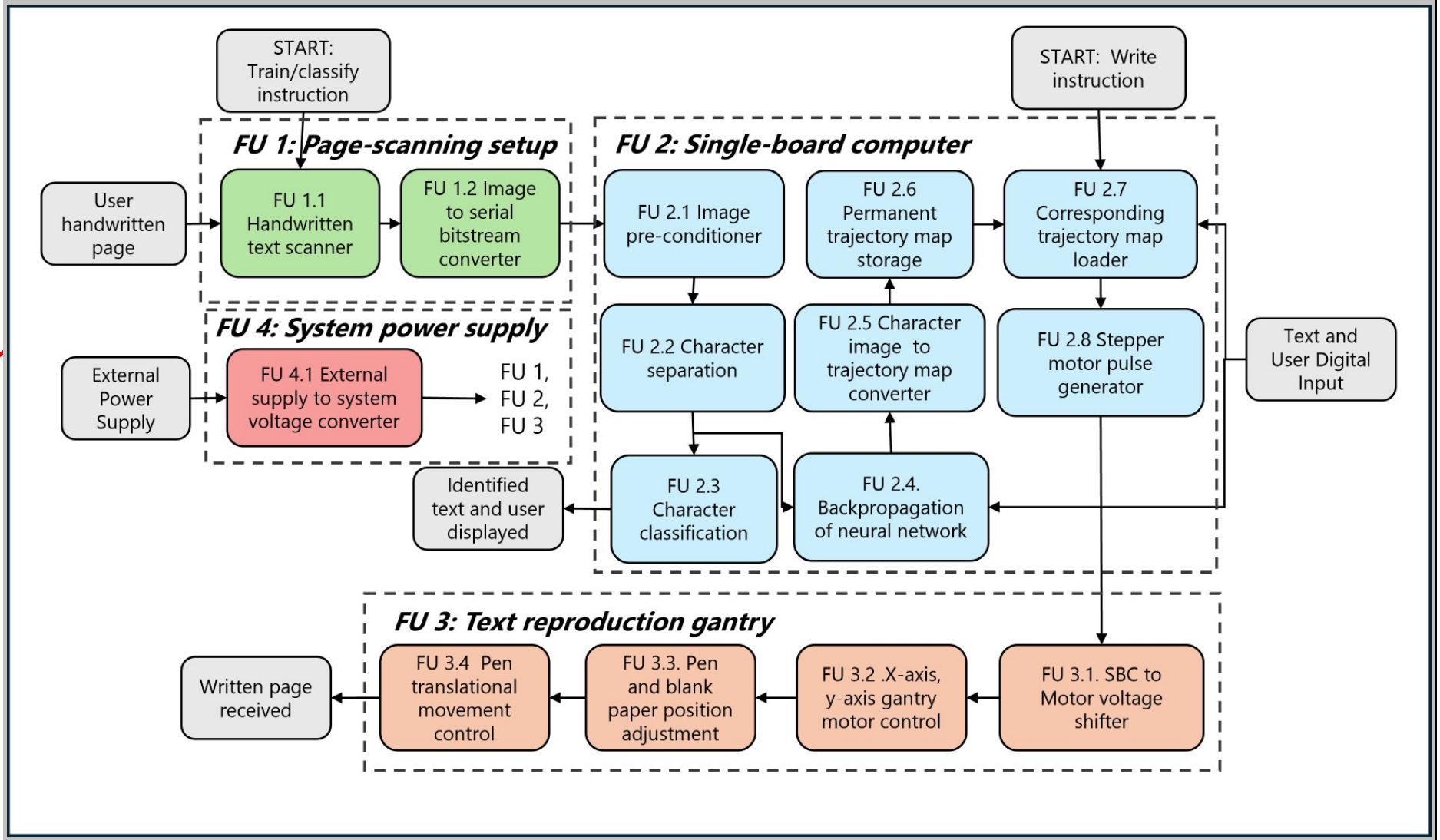
There are three main modes of operation for the handwriting robot: training, classification and reproduction

Training/Classification: First, the user will input their page of handwritten text to the scanner (FU 1.1). The user will select to either train or classify, and will input the intended digital text alongside this through an input interface. Once the page is scanned, a camera USB driver will be used to transfer the image to the single board computer (FU 1.2). The image will be conditioned first to fix warping or noise in the image (FU 2.1), and then the character separation will occur. This will separate the page characters (or groups of characters that are connected) to match the desired neural network input size (FU 2.2). The difference between training and classification occurs here: if the 'training' option is selected, the separated characters will be used to adjust the neural network neuron weightings against the desired digital text that was input (FU 2.4). Each separated character or group of characters will also be converted into a trajectory map (coordinate map of required pen motion) (FU 2.5). This trajectory map will be stored permanently within the system (FU 2.6). If the 'classify' option is selected, the separated characters or group of characters will be passed into the trained neural network classifier (FU 2.3), and the corresponding digital text and detected user will be outputted to the display interface.

Reproduction: If the user sends a 'write' instruction, the digital text and user style desired to be reproduced will be provided. The corresponding trajectory for each character/group will be retrieved from storage (FU 2.7), and will be converted into stepper motor controls required to produce the desired pen movements (FU 2.8). These stepper motor controls will be sent to the corresponding motors via a stepper motor driver to allow for forward and reverse movements (FU 3.1). The x-motor, y-motor and pen lift mechanism will be controlled through the stepper motor driver (FU 3.2/3.4). After the pen has been adjusted and is held securely by the pen holder in the robot (FU 3.3), precise handwritten imitations of each character will be written.

Remove power supply (FBD should shown system functions).
Remove the "start" blocks - this is not a flow diagram, but rather an FBD.

## 3.2 Functional block diagram (this should not be a flow diagram)



## 4. System requirements and specifications

These are the core requirements of the system or product (the mission-critical requirements) in table format **IN ORDER OF IMPORTANCE**. Requirement 1 is the most fundamental requirement.

| | **Requirement 1: the fundamental functional and performance requirement of your project** | **Requirement 2** (Second in order of importance) | **Requirement 3** (Third in order of importance) |
|---|---|---|---|
| **1. Core mission requirements of the system or product.** Focus on requirements that are **core** to solving the engineering problem. These will reflect the solution to the problem. | The system must be able to accurately reproduce any desired digital text on paper in the writing style of any writer from the training set. | The system must be able to accurately decode the content of the handwritten text, and identify which writer from the training set wrote the handwritten text. | The system must reproduce text on paper at a near real time speed, at a rate that can be considered practical and useful. |
| **2. What is the target specification** (this must be in *measurable* terms) to be met in order to achieve the requirement in 1. above? | The system should reproduce desired text with a minimum character accuracy of 85% while having a high stylistic correlation of 85% to the chosen writer's style. | The system must be able to identify the writer of the handwritten text from the training set and the content of the written text with an overall accuracy of 95%. | The system must be capable of reproducing the desired text in the writing style of any writer in the training set at a rate of no slower than 4 seconds per character. |
| **3. Explain where this specification originated :** i.e., **explain** the value that you selected. I.e., *how* will meeting the specification given in point 2 above *solve the problem*? | An accuracy of 85% (averaged from Babushkin et al. results [1, p8] ) is high enough to represent the input text, ~~but practical to account for the build up of precision loss in the system.~~ see class discussion on what specs should be based on. | An overall classification accuracy of 95% (similar to existing classifiers mentioned in Acta Informatica Pragensia [2, p6] ) ensures practical usability and consistency. | Averaging the time required to produce text between 0% and 1% error (Ossi Teikari [3, p35]) gives approximately 4 seconds per character, allowing the system to reproduce text at a practical rate with minimal error. |
| **4. How will you demonstrate at the examination** that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will ***prove*** these were met. | A sentence of text will be reproduced in a specific user style, and the amount of characters that can be visibly seen to match the input text (in the style of the chosen user) must be >= to 85% of the total characters produced. This is not a proper way to measure. write in text | A predefined paragraph of text must be handwritten by a writer in the training set and classified by the system. The classified text and writer must match the input text and writer with an overall accuracy of >= 95% . | A sentence of x characters long will be sent to the system to be reproduced, the system must be able to reproduce it within 4*x seconds. this means "convolution" |
| **5. Your own design contribution:** what are the aspects that *you will design and implement yourself* to meet the requirement in point 2? If none, *remove this requirement.* | Accurate conversion from character image to pen trajectory map, precise motor control from pen trajectory map, and a structurally sound gantry design for accurate drawing will be designed and implemented by the student. | A rig to capture an image of written text, image conditioning and noise removal, character separation and processing, classification with a neural network, and communicating necessary data to a display will be designed by the student. | Techniques to optimize pen movements for quicker drawing, smooth motor control and robust design for efficient gantry movements will be designed and implemented by the student. |
| **6. What are the aspects to be taken off the shelf** to meet this requirement? If none, indicate "none".<br>Clearly specify for what tasks **library functions** will be used (if relevant to the project). | Physical motors, pen, single board computer, stepper motor driver, and separate components for gantry design e.g. screws or belts will be taken off the shelf. | A camera, camera driver (library function), single board computer, separate components for image-capturing rig, and a physical device where data is displayed will be taken off the shelf. | Efficient physical motors, single board computer, stepper motor driver, and separate components for gantry design e.g. smooth ball bearings will be taken off the shelf. |

the NUMBER of characters, NOT amount.

## System requirements and specifications page 2

| | **Requirement 4** (Fourth in order of importance) | **Requirement 5** | **Requirement 6** |
|---|---|---|---|
| **1. Core mission requirements of the system or product.** Focus on requirements that are core to solving the engineering problem. These will reflect the solution to the problem. | The system must be able to identify the writer of the handwritten text, and decode the content of the handwritten text in a short, practical amount of time. | The gantry of the system must have precise pen movements and accurately draw the desired characters with minimal positional error. | The system should be able to be trained on multiple different writing styles and be capable of storing all the writers' necessary character trajectory maps within the system permanently. |
| **2. What is the target specification** (this must be in *measurable* terms) to be met in order to achieve the requirement in 1. above? | The system must identify the writer of the written text from the training set and decode the content of the text at a rate of no slower than 25ms per character with an extra 2 seconds for image capture/conditioning. | The gantry of the system must be capable of drawing the desired character trajectory map while maintaining an absolute positional error that does not exceed 0.5mm. 0.5 mm relative to text that is how large? | The system should be capable of training on a minimum of 10 different users' writing styles, and be capable of storing the necessary character mappings for each of these 10 user's permanently. |
| **3. Explain where this specification originated:** i.e., explain the value that you selected. I.e., *how* will meeting the specification given in point 2 above solve the problem? | 2 seconds was given to image capturing and conditioning due to transmission times and complex conditioning and noise removal algorithms, and 25ms per character classification due to simple forward propagation calculations. same comment as before. Specs are selected to best solve the problem, not because "processing will take long". | According to research by T. Huang and R. Xiong [4] an absolute positional error of 0.3mm with a gantry is achievable and indicates sufficient accuracy, therefore a maximum absolute error of 0.5mm was chosen for slightly more leeway. imcorrect motivation | Having 10 trained user styles stored at the same time proves multi-user neural network recognition and multi-user storage. More than 10 is not required due to neural network complexity and storage limitations. same comment again |
| **4. How will you demonstrate at the examination** that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will *prove* these were met. | A page with a sentence of x handwritten characters will be given to the system. The time from the initial instruction to the received classified text and user should be within 2 + x*0.025 seconds. | The system should be instructed to recreate the same word twice without varying the dataset characters, and the two words should appear identical on-top of each other, and not exceed an absolute positional error of 0.5mm. | The system should display the current amount of users in the system, and any random user should be selected out of the 10 and have text reproduced in their style to prove the system can be trained with 10 different users. |
| **5. Your own design contribution:** what are the aspects that *you will design and implement yourself* to meet the requirement in point 2? If none, *remove this requirement.* | A rig to capture an image of written text, image conditioning,optimized and efficient character separation, neural network character classifier, and calculation and display of time taken will be designed and implemented by the student. | Accurate conversion from pen trajectory map to precise stepper motor control, robust gantry system to write characters, secure pen adjuster and holder, and a secure paper holder will be designed and implemented by the student. | A rig with a camera to capture new text, sufficiently complex neural network to be able to classify minimum 10 user's styles, and a storage strategy to store 10 user's character trajectory mappings will be designed by the student. |
| **6. What are the aspects to be taken off the shelf** to meet this requirement? If none, indicate "none".<br>Clearly specify for what tasks **library functions** will be used (if relevant to the project). | A camera, camera driver, single board computer, separate components for image-capturing rig, and a physical device where time is displayed will be taken off the shelf. | Precise stepper motors, separate components needed for gantry, pen and paper holders, and a pen will be taken off the shelf. This show repeatability, not accuracy | A camera, camera driver, single board computer, components for image-capturing rig, and permanent storage device will be taken off the shelf. NUMBER of users. Not amount. |

## System requirements and specifications page 3

| | **Requirement 7** (Seventh in the order of importance) | **Requirement 8** | **Requirement 9** |
|---|---|---|---|
| **1. Core mission requirements of the system or product.** Focus on requirements that are core to solving the engineering problem. These will reflect the solution to the problem. | | | |
| **2. What is the target** specification (this must be in *measurable* terms) to be met in order to achieve the requirement in 1. above? | | | |
| 3. **Explain where this specification originated**: i.e., explain the value that you selected. I.e., *how* will meeting the specification given in point 2 above solve the problem? | | | |
| 4. **How will you demonstrate at the examination** that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will *prove* these were met. | | | |
| **5. Your own design contribution:** what are the aspects that *you will design and implement yourself* to meet the requirement in point 2? If none, *remove this requirement.* | | | |
| **6. What are the aspects to be taken off the shelf** to meet this requirement? If none, indicate "none". Clearly specify for what tasks l**ibrary functions** will be used (if relevant to the project). | | | |

## 5. Real world field conditions

These are the ACTUAL, REAL WORLD CONDITIONS under which your project has to work *and will be demonstrated.*

| | **Real world field condition 1** (required to include) | **Real world field condition 2** (optional to include) | **Real world field condition 3** (optional to include) |
|---|---|---|---|
| **Field condition requirement.** In which real world conditions does the system have to operate?E.g., "lab conditions" or "outside in uncontrolled conditions". Describe the one, two or three most important field conditions. | The system is expected to perform in well-lit areas like a lab or office environment, at an operating range of 400-1000 lux, to ensure minimal noise in image capturing. | The system is expected to run on a stable, level and vibration free surface, to ensure the page scanning is accurate and the pen in the gantry can write precisely without fluctuations. | |

## 6. Student tasks

### 6.1 Design and implementation tasks of the student

List your primary design and implementation tasks in bullet list format (5-10 bullets). These are *NOT* product requirements, but *YOUR* tasks.

- The student will research types of neural networks, their structure, the effect of neurons and layers, and how to extract features of text from different layers.
- The student will research image conditioning, noise removal, character separation strategies, and pen trajectory mapping.
- The student will design and implement the convolution neural network, image conditioning, and character separation algorithms.
- The student will simulate the classifier on existing handwritten text datasets with adjusted character separation strategies and neuron/layer numbers to meet requirements.
- The student will construct a rig to capture images of handwritten text on an A4 page, and implement the neural network and algorithms on a chosen single board computer.
- The student will test the new classifier implementations on the single board computer with the captured images to ensure requirements are met.
- The student will design the gantry system in CAD software and construct it with the necessary store-bought individual components.
- The student will design and implement the pen trajectory mapping algorithm and the stepper motor control.
- The student will test the constructed gantry with the implemented classifier to ensure the remainder of the requirements are met.

### 6.2 New knowledge to be acquired by the student

Describe what the theoretical foundation to the project is, and which *new* knowledge you will acquire (*beyond* that covered in any other undergraduate modules).

- The student will acquire new knowledge of training and classification of character images using a neural network.
- The student will acquire new knowledge involving the extraction of text features from neural network layers.
- The student will acquire new knowledge about the process of image processing with character isolation.
- The student will acquire new knowledge about pen trajectory map creation.
- The student will acquire new knowledge on 2D gantry design and implementation with precise motor control.
- The student will acquire new knowledge about the implementation of neural networks on a single board computer.

## 7. References

Format your reference list either in IEEE reference format or in APA reference format. Check what your study leader prefers and use consistently.


[1] V. Babushkin, H. Alsuradi, M. O. Al-Khalil, and M. Eid, “Analyzing handwriting legibility through hand kinematics,” Frontiers in Artificial Intelligence, vol. 8, Mar. 2025, doi: https://doi.org/10.3389/frai.2025.1426455.

[2] Sidar Agduk and Emrah Aydemir, “Classification of Handwritten Text Signatures by Person and Gender: A Comparative Study of Transfer Learning Methods,” Acta Informatica Pragensia, vol. 11, no. 3, pp. 324–347, Dec. 2022, doi: https://doi.org/10.18267/j.aip.197.

[3] O. Teikari, "Reproducing Handwriting with Position-Controlled Robots," Master's thesis, Dept. Pervasive Systems, Tampere University of Technology, Tampere, Finland, 2016.

[4] T. Huang and R. Xiong, "Cost-Effective Robotic Handwriting System with AI Integration," in Proc. 2024 IEEE Long Island Systems, Applications and Technology Conference (LISAT), 2024.